curve_traits: the last 17 open End(E) levels, by descending isogeny walks

2026-10-28. Code: `src/cryptanalysis/curve_traits/walk.rs` (`height_above_floor`) and `mod.rs` (`settle_by_descent`). Output: `docs/curves/traits.json`.

The open levels

17 registry curves over 47- and 53-bit prime fields, added after the earlier level work, had one open volcano level each:

- the 2-volcano for $d_K = -3, -4, -7$, depth 20 or 24;
- the 3-volcano for $d_K = -3$, depth 12.

The prime-power scalar test and the horizontal-isogeny cycles had bounded each level at 1 or 3.

These bounds could not be closed by the cycles. A cycle of two 11-isogenies fits both level 2 ($h(\mathcal{O}_4) = 2$) and level 3 ($h(\mathcal{O}_8) = 4$), since the class above 11 has order 2 in both.

Method

With $\ell \mid v$, every vertex of the ℓ-volcano above the floor has $\ell + 1$ rational ℓ-isogenies, and a floor vertex has exactly one. Below the crater one edge goes up and ℓ go down. So a walk on j-invariants that never steps back along the edge it arrived by keeps descending once it has.

From each rational neighbour of $j(E)$ (the roots of $\Phi_{\ell(j,Y)}$ in $\mathrm{GF}(p)$), the walk runs until it reaches a vertex with a single rational ℓ-isogeny. The shortest such walk has length exactly the height h above the floor, and the level is depth $- h$ (Kohel; Fouquet–Morain). Φ_2 was added to the committed modular polynomials. A level above the bound the other tests reached stops the build.

Checks

- **Unit test.** Heights match a breadth-first search of the Φ_ℓ graph in PARI/GP (`bfs.gp`, `gen.gp`, `vectors.txt`) on 40 random curves over 17- to 20-bit prime fields, covering $\ell = 2, 3, 5$ and levels 0 to 6.
- **The 17 curves.** An independent PARI/GP implementation of the walks (`walk.gp`, `pari_levels.txt`) gives the same 17 levels (`levels.json`).
- **Consistency with the cycles.** Every level is within the earlier bound. The class order of the prime above 11 in $\mathrm{Cl}(\mathcal{O}_f)$ agrees with each closed 11-isogeny cycle.

Result

	before	after
End(E) proved	19,876	19,893
End(E) bounded (open levels)	17	0

Only the 17 records change, with 0 regressions. No volcano level is open in the catalogue.

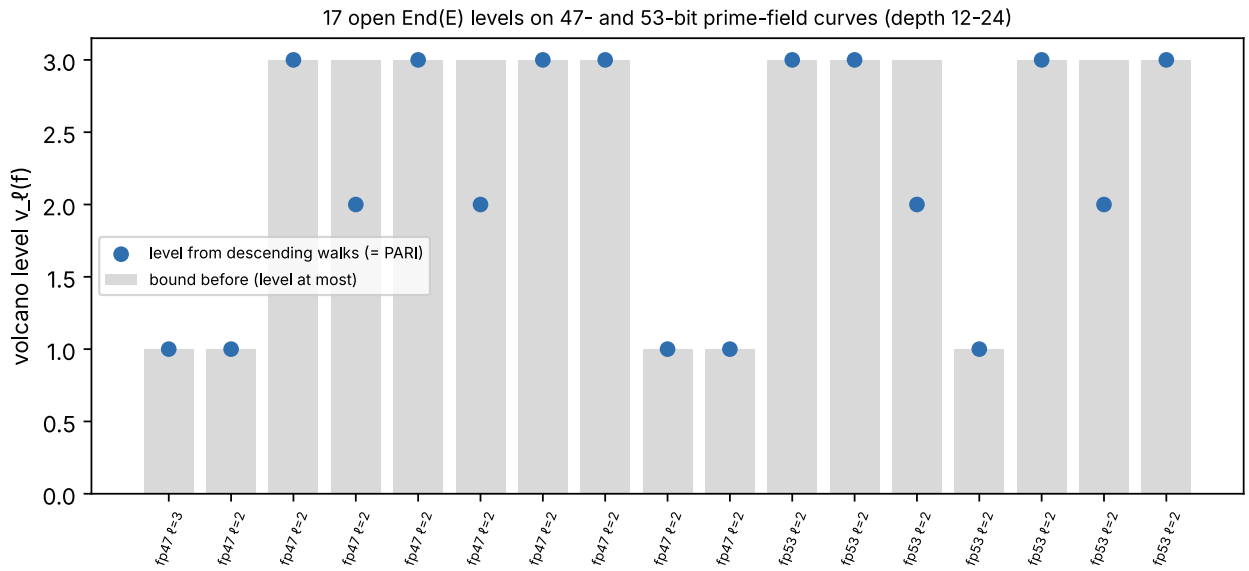


Figure 1: Bound before and level after, per open level.